ECON 0100 | HW E2 | Competitive Firms

Solutions

Answers and work are shown in the red blocks.

Q1 | A Price for Brooms

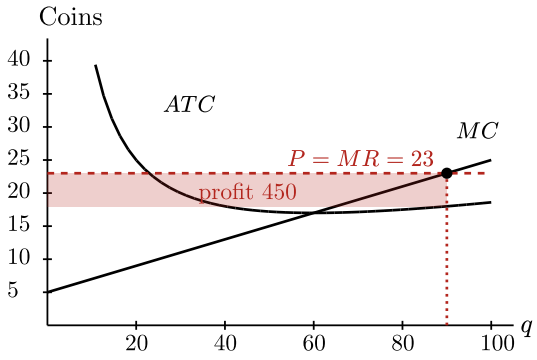
In the early days after the discovery of autolocomotion, SwiftTech's method for broomstick manufacturing spread widely. Many identical makers began selling the same standard broom, each one taking the market price. Each maker's fixed cost is 360 coins a week, with a marginal cost of:

$$MC = 5 + \frac{q}{5}$$

where q is the number of brooms it makes in a week. The table shows its average total cost at some quantities. Each graph shows one maker's marginal cost and average total cost.

Brooms	**20**	**40**	**60**	**90**
ATC	25	18	17	18

If the market price is 23 coins:



a) Find and graph marginal revenue (MR).

Marginal revenue is the price, 23 coins: a flat line at 23.

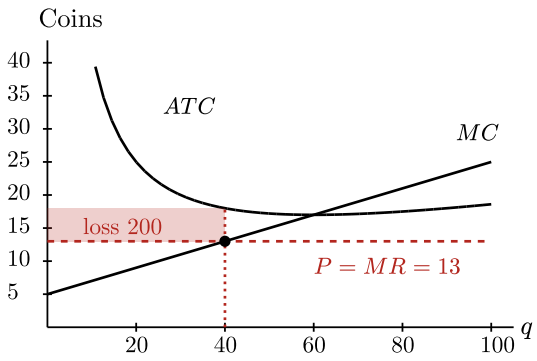
b) Find and graph the firm's best quantity (q^*).

$P = MC$: $23 = 5 + q/5$, so $q^* = 90$ brooms.

c) Find and graph the firm's maximized profit.

At 90 brooms ATC is 18, so profit is $(23 - 18) \times 90 = 450$ coins: the rectangle between the price line and ATC, out to 90.

If the price falls to 13 coins:



d) Find and graph the firm's q^* if they stay open.

$13 = 5 + q/5$, so $q^* = 40$ brooms.

e) Find and graph the firm's maximized profit if open.

At 40 brooms ATC is 18, so profit is $(13 - 18) \times 40 = -200$ coins: a loss of 200, the rectangle between ATC and the price line, out to 40.

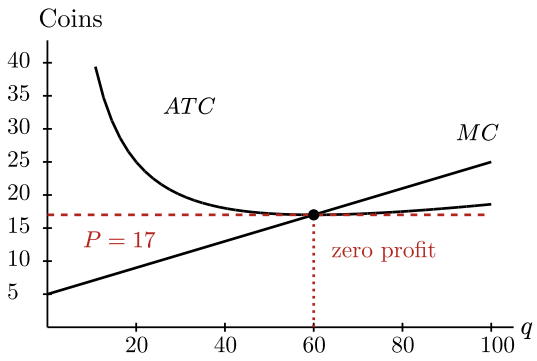
f) What is the firm's profit / loss if they shutdown?

It still pays the fixed cost and sells nothing: a loss of 360 coins.

g) Should the firm stay open or shutdown?

Stay open. It loses 200 instead of 360, because the price of 13 more than covers the variable cost of each broom.

In the long run:



h) Do firms enter or exit at a price of 23 coins?

> Firms enter. At 23 each firm earns a profit, which draws new firms in and pushes the price down.

i) Do firms enter or exit at a price of 13 coins?

> Firms exit. At 13 each firm takes a loss, so firms leave and the price rises.

j) Find and graph the long-run price.

> Entry and exit stop when profit is zero, where the price equals the lowest ATC: 17 coins.

k) Find and graph the firm's q^* and profit at that price.

> $17 = 5 + q/5$, so $q^* = 60$ brooms. Profit is $(17 - 17) \times 60 = 0$: the price line just touches ATC, so there is no rectangle.

Q2 | Subsidizing Broomsticks

Quidditch is a game much beloved by the magical community. Since the rules are well established, the game equipment is standardized, and every equipment seller produces identical equipment. Broomsticks are an important part of the game, and anyone who wants to can start making and selling them. The number of sellers of broomsticks hasn't changed much in the over 500 years since SwiftTech's methods spread.

To spur interest in the sport, the Ministry imposed a subsidy on broomsticks. Using a couple of graphs to illustrate your answer, explain what happened in the market because of this subsidy. Be sure to include all relevant components on the graph, and conclude in a long-run equilibrium.

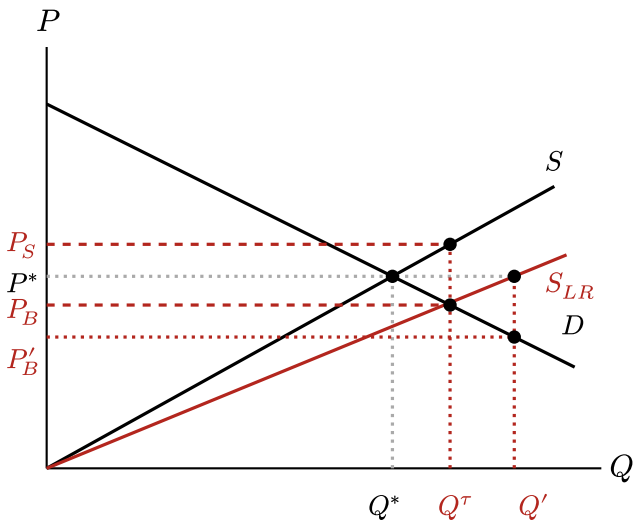
> **1. Before the subsidy.** Broomsticks are identical, anyone can enter, and the number of sellers has been steady, so the market starts in long-run equilibrium. The price P^* sits at the lowest point

of each seller's ATC, where $MR = P = ATC = MC$. Each seller makes q^*, the market trades Q^*, and profit is zero.

2. Short run. The subsidy puts a gap between the price sellers receive, P_S, and the price buyers pay, P_B. The market quantity rises to Q^τ. Each seller's marginal revenue is now P_S, above P^*, so it makes more ($q^\tau > q^*$) and earns a profit, because P_S is above its ATC. Profit leads to entry.

3. Long run. New sellers shift market supply out, and the price sellers receive falls. Entry stops when profit is zero again, which is when sellers are back to receiving $P'_S = P^*$. Buyers then pay less by the full subsidy, P'_B. Each seller is back to $q' = q^* < q^\tau$, there are more sellers, and the market quantity is larger still: $Q' > Q^\tau > Q^*$.

The Market



One Firm

